

Question Paper

Exam Date & Time: 16-Jun-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

FOURTH SEMESTER B.TECH. (COMPUTER SCIENCE) DEGREE EXAMINATIONS - JUNE 2025
SUBJECT: CSE 2224 / CSE_2251 - DATABASE SYSTEMS
(MAKE-UP)

Marks: 50

Duration: 180 mins.

Answer all the questions.

- 1A) Differentiate between various database users (end users, database administrators, system analysts) in terms of roles and responsibilities. (2)
- 1B) Product(PID, PName, Price, Category) (6)
Customer(CID, CName, Email, City)
Orders(OID, CID, PID, Quantity, OrderDate)
Based on the above schemas, write the SQL queries for the following:
i) Find the product name and quantity ordered by the customer named 'XYZ'.
ii) List the names of customers who have ordered at least one product from every category.
iii) Find the names of customers who have never placed an order.
iv) List the customers who have placed more than 5 orders in total.
Based on the above schemas, write the SQL queries for the following:
i) Find the product names that have not been ordered by any customer from the city 'ABC'.
ii) Find the names of products that have the highest total quantity ordered.
(4+2 = 6 marks)
- 1C) In the Timestamp Ordering protocol, when a transaction is rolled back, it is assigned a new timestamp rather than retaining its original one. Why is it necessary to assign a new timestamp instead of reusing the old one? Additionally, if we used the wording "write(Q)" instead of "w-timestamp(Q)" in the protocol, would this change have any impact on its behaviour or correctness? Justify your answer. (2)
- 2A) Two transactions are running concurrently. The schedule is as follows: (6)
T1: R(A), W(A), R(B), W(B), COMMIT
T2: R(B), W(B), R(A), W(A), COMMIT
Assuming strict 2PL and that each read/write requires an exclusive lock:
i) List the lock and unlock operations.
ii) Show whether any deadlock occurs. If deadlock occurs, which transaction should be aborted?
(3+3 = 6 marks)
- 2B) Design an Entity-Relationship Diagram for an E-Commerce System where entities include (4)
Customer, Order, Product, Payment, and Shipment.
Include the following in ER diagram:
i) Define the primary keys for each entity.
ii) Specify relationships such as *places* between Customer and Order, *contains* between Order and Product.
iii) Include cardinality (e.g., one-to-many between Customer and Order).
iv) Address participation constraints (e.g., every order must include at least one product)
- 3A) Consider a relation schema R(A, B, C, D, E) with the following set of functional dependencies: (5)
 $A \rightarrow B$
 $A \rightarrow B C$

$A \rightarrow B \rightarrow D$

$A \rightarrow D \rightarrow E$

i) Find the canonical cover for the functional dependencies.

ii) Find the candidate key for this R.

iii) Decompose the relation to 3NF.

3B) Given $R(A, B, C, D, E)$ with the following (5)

dependencies: $\{AB \rightarrow CD, D \rightarrow E, A \rightarrow C, B \rightarrow D\}$.

Decompose the relation into BCNF.

4A) Construct a B+ tree for the following set of key values with $n=4$: (5)

5 10 15 25 30 35

Show the form of the tree after each of the following operations:

i) Delete 10

ii) Delete 35

4B) A file contains records with the following search key values: 6,7,8,14,18,19,22,23,24,25,27,30,49. (3)

Show the extendable hash structure for this file if the hash function $h(x) = x \bmod 7$ and the buckets can hold three records.

4C) With neat diagrams, show the initial File with fixed length records and show different techniques for (2)
deleting records in this file.

5A) Describe first normal form? Analyze if the Table 1 is in 1NF? Justify your answer. (3)

Table1: Student details

Student_ID	StudentName	Course
502	Alice	Math
567	Chuck	Physics
789	Bob	Chemistry
899	Chuck	Biology

5B) Consider the precedence graph of Figure 5B. Is the corresponding schedule conflict serializable? (2)
Justify your answer.

i)

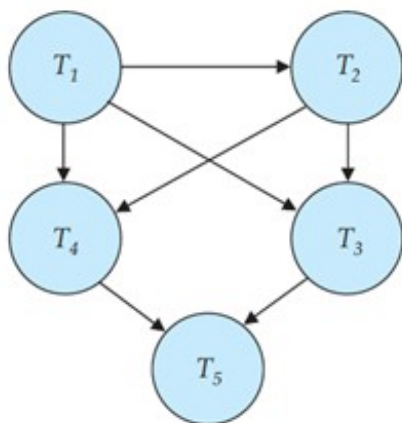


Figure 5B. Precedence graph

ii) "Concurrent execution of transactions is more important when data must be fetched from (slow) (2)
disk or when transactions are long, and is less important when data are in memory and transactions
are very short" - Is it True? Justify your answers.

5C) Justify the reason as to why log records for transactions on the Undo-list must be processed in (3)
reverse order, whereas Redo-list is performed in a forward direction - illustrate with suitable
examples.

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Event Name	Subject
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#	Course Outcome	Description	Marks
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No records found / CO Level Description not defined

* **NA** - CO Not Selected

Event Name	Subject
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MIT MPL and BLR- BTech-MSc-MCA - II-IV and VI Sem - Makeup Examination - Jun 2025

CSE-2251:Database Systems

#	Level	No of Questions	Marks (%)
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